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Electric cylinder and work machine

Abstract

An electric cylinder includes a motor that is a driving source, an output shaft configured to rotate in response to driving of the motor, a sun gear coupled to the output shaft and configured to rotate in response to the rotation of the output shaft, a tubular holder disposed adjacent to an axial end surface of the motor and including a fluid accommodation portion that allows a magnetic fluid to be accommodated between the holder and an outer periphery of the sun gear, a pair of magnetic pole pieces provided on an inner peripheral surface of the holder, and a magnet disposed between the pair of magnetic pole pieces and configured to hold the magnetic fluid in the fluid accommodation portion by using an action of a magnetic field.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an electric cylinder and a work machine.
(2) Priority is claimed on Japanese Patent Application No. 2022-007049, filed Jan. 20, 2022, the content of which is incorporated herein by reference.

BACKGROUND ART

(3) Patent Document 1 discloses an electric excavator as an example of a work machine. The

electric excavator includes a vehicle body, a boom that is provided to be rotationally movable with respect to the vehicle body, and an electric cylinder. The boom is driven by the electric cylinder.

(4) On the other hand, the electric cylinder may include a planetary gear mechanism that transmits the driving force of a motor, which is a driving source, to a piston. The planetary gear mechanism includes a sun gear coupled to an output shaft of the motor, and a plurality of planetary gears adjacent to the sun gear. The sun gear rotates in response to the rotation of the output shaft. The plurality of planetary gears rotate in response to the rotation of the sun gear. A tooth surface of the sun gear is in contact with a tooth surface of the planetary gear.

CITATION LIST

Patent Document

(5) [Patent Document 1]

(6) Japanese Unexamined Patent Application, First Publication No. 2020-204172

SUMMARY OF INVENTION

Technical Problem

(7) In a case of Patent Document 1, the boom rotationally moves in response to driving of the electric cylinder. In Patent Document 1, in order to prevent a problem caused by heat generation of the motor which is the driving source of the electric cylinder, it is required to efficiently release heat from the motor to the outside. For example, in a case in which the motor serving as power is downsized, a proportion of the motor used in an acceleration region is increased, and thus heat release from a motor surface alone is not sufficient, and the temperature is likely to increase. In a case in which the motor generates excessive heat, the performance of the motor may deteriorate or the motor may malfunction. Therefore, it is necessary to efficiently transfer the heat of the motor and release the heat to the outside.

(8) On the other hand, in a case in which the electric cylinder includes the planetary gear mechanism, a peripheral portion of the sun gear is heated to a high temperature due to the heat transferred from the output shaft of the motor and the heat generated on the tooth surface of the sun gear. In this case, a lubricant, such as grease, may melt out due to the high temperature, and the failure of the lubrication is likely to occur. In a case in which the failure of the lubrication occurs, the gear surface is locally deteriorated or worn due to frictional heat, which causes a problem such as a sliding portion of the sun gear to be stuck. Therefore, in order to prevent the problem caused by the temperature rise of the peripheral portion of the sun gear, it is required to efficiently release the heat of the sun gear tooth surface to the outside along with the heat from the motor.

(9) An object of the present invention is to provide an electric cylinder and a work machine that can efficiently release heat from a motor and heat from a tooth surface of a sun gear to the outside.

Solution to Problem

(10) An electric cylinder according to an aspect of the present invention includes: a motor that is a driving source; an output shaft configured to rotate in response to driving of the motor; a sun gear coupled to the output shaft and configured to rotate in response to the rotation of the output shaft; a tubular holder disposed adjacent to an axial end surface of the motor and including a fluid accommodation portion that allows a magnetic fluid to be accommodated between the holder and an outer periphery of the sun gear; a pair of magnetic pole pieces provided on an inner peripheral surface of the holder; and a magnet disposed between the pair of magnetic pole pieces and configured to hold the magnetic fluid in the fluid accommodation portion by using an action of a magnetic field.

Advantageous Effects of Invention

(11) According to the above-described aspect, the heat from the motor and the heat from the tooth surface of the sun gear can be efficiently released to the outside.

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view of an excavator according to an embodiment.
- (2) FIG. 2 is a side view of work equipment according to the embodiment, which shows an inside of a boom and an arm in a transparent manner.
- (3) FIG. 3 is a perspective view of the boom according to the embodiment, which shows the inside of the boom in a transparent manner.
- (4) FIG. 4 is a perspective view of the arm according to the embodiment, which shows the inside of the arm in a transparent manner.
- (5) FIG. 5 is a perspective view of an electric cylinder according to the embodiment.
- (6) FIG. 6 is a view of the electric cylinder according to the embodiment when viewed from one side in an axial direction.
- (7) FIG. 7 is a view of the electric cylinder according to the embodiment when viewed from the other side in the axial direction.
- (8) FIG. 8 is a view including a cross section of VIII-VIII of FIG. 7.
- (9) FIG. 9 is an enlarged view of a portion of IX of FIG. 8.
- (10) FIG. 10 is a view including a cross section of X-X of FIG. 9.
- (11) FIG. 11 is a perspective view of a first carrier according to the embodiment.
- (12) FIG. 12 is a perspective view of a second carrier according to the embodiment.
- (13) FIG. 13 is an exploded perspective view of a peripheral structure including a fluid holding structure according to the embodiment.
- (14) FIG. 14 is a cross-sectional perspective view of the peripheral structure including the fluid holding structure according to the embodiment.
- (15) FIG. 15 is a view of the peripheral structure including the fluid holding structure according to the embodiment when viewed from one side in the axial direction.
- (16) FIG. 16 is a view of the peripheral structure including the fluid holding structure according to the embodiment when viewed from the other side in the axial direction.
- (17) FIG. 17 is a view including a cross section of XVII-XVII of FIG. 16.
- (18) FIG. 18 is a view showing an exemplary example of a flow of a lubricant according to the embodiment.
- (19) FIG. 19 is a view showing an exemplary example in which heat from a motor according to the embodiment and heat from a tooth surface of a sun gear are moved.

DESCRIPTION OF EMBODIMENTS

- (20) Hereinafter, an embodiment of the present invention will be described with reference to the drawings. In the embodiment, an excavator will be described as an exemplary example of a work machine (work vehicle).
- (21) <Excavator (Work Machine)>
- (22) As shown in FIG. 1, an excavator 1 as the work machine includes a vehicle body 2 and work equipment 3 coupled to the vehicle body 2. Hereinafter, an advance direction, a retraction direction, and a vehicle width direction of the excavator 1 will be referred to as a “front side of a vehicle (one side in a front-rear direction of the vehicle)”, a “rear side of a vehicle (the other side in the front-rear direction of the vehicle)”, and a “vehicle width direction”. The vehicle width direction may be referred to as a “left side (one side in the vehicle width direction)” or a “right side (the other side in the vehicle width direction)”. A right hand with respect to the advance direction of the excavator 1 will be referred to as a right side, and a left hand with respect to the advance direction of the excavator 1 will be referred to as a left side. An up-down direction, an upper side, and a lower side of the excavator 1 in a state in which the excavator 1 is disposed in a horizontal plane will be simply referred to as an “up-down direction”, an “upper side”, and a “lower side”.
- (23) <Vehicle Body>
- (24) The vehicle body 2 includes an autonomous undercarriage 5 and an upper swing body 6 that is

provided on the undercarriage 5 to be able to swing.

(25) The undercarriage 5 includes a pair of left and right crawler belts 7. The undercarriage 5 includes an electric motor (not shown) that drives the crawler belts 7. The undercarriage 5 travels by driving the crawler belts 7 via the electric motor. It should be noted that the undercarriage 5 may be provided with a hydraulic motor instead of the electric motor.

(26) A blade 8 as a dirt discharging plate extending in the vehicle width direction of the undercarriage 5 is provided at a front portion of the undercarriage 5. The undercarriage 5 includes an electric actuator (not shown) that drives the blade 8. The height position of the blade 8 can be adjusted by driving the electric actuator.

(27) The upper swing body 6 is provided above the undercarriage 5. The upper swing body 6 is provided with a traveling electric motor that is a driving source of the undercarriage 5, a motor of an electric cylinder that is a driving source of the work equipment 3, a battery serving as a power source of each motor, an inverter, and the like (not shown). The upper swing body 6 is configured to swing about an axis extending in the up-down direction with respect to the undercarriage 5.

(28) A canopy 10 is provided on the upper swing body 6. The canopy 10 has a driving space 11 in which a driver can be accommodated. The canopy 10 includes a hood 12 that forms a ceiling portion of the driving space 11, rear portion pillars 13 that are provided on both sides of the hood 12 in the vehicle width direction at a rear portion of the hood 12 and that extend downward from the hood 12, and front portion pillars 14 that are provided on both sides of the hood 12 in the vehicle width direction at a front portion of the hood 12 and that extend downward from the hood 12.

(29) A bracket 15 that supports a boom 20 is provided on a front portion of the upper swing body 6. As shown in FIG. 2, the bracket 15 includes a first hole 15a and a second hole 15b that are open in the vehicle width direction of the upper swing body 6. The first hole 15a is disposed in the vicinity of an upper end portion of the bracket 15. The second hole 15b is disposed below and in front of the first hole 15a.

(30) <Work Equipment>

(31) As shown in FIG. 1, the work equipment 3 is provided to be operable in a bending and undulating manner with respect to the upper swing body 6. The work equipment 3 includes the boom 20, an arm 30, a bucket 40 (work tool), and a plurality (for example, three in the present embodiment) of electric cylinders 100A to 100C. The three electric cylinders 100A to 100C are a first electric cylinder 100A that operates the boom 20, a second electric cylinder 100B that operates the arm 30, and a third electric cylinder 100C that operates the bucket 40. A base end portion of the boom 20 is rotatably coupled to the upper swing body 6. A distal end portion of the boom 20 is rotatably coupled to a base end portion of the arm 30. A distal end portion of the arm 30 is rotatably coupled to the bucket 40.

(32) <Boom>

(33) In a posture of FIG. 1, the boom 20 extends upward from the bracket 15 and then is bent and extends forward and upward when viewed from the vehicle width direction of the upper swing body 6. Hereinafter, when viewed from the vehicle width direction of the upper swing body 6, a direction in which the boom 20 extends will be referred to as a “boom extension direction”, and a direction orthogonal to the boom extension direction will be referred to as a “boom plate width direction”. One end portion (end portion on the bracket 15 side) of the boom 20 in the boom extension direction will be referred to as a “boom base end portion”. Another end portion (end portion on a side opposite to the boom base end portion) of the boom 20 in the boom extension direction will be referred to as a “boom distal end portion”. The dimension in the boom plate width direction gradually increases from the boom base end portion toward the vicinity of the center in the boom extension direction, and then gradually decreases toward the boom distal end portion.

(34) As shown in FIG. 3, the boom 20 includes a pair of boom side plates 21 that are disposed to be spaced apart from each other in the vehicle width direction of the upper swing body 6, a boom

bottom plate **22** that extends in the vehicle width direction of the upper swing body **6** to connect the pair of boom side plates **21** to each other, a boom base end side connection plate **23** connected to the boom base end portion side of the boom bottom plate **22**, a boom distal end side connection plate **24** connected to the boom distal end portion side of the boom bottom plate **22**, a boom partition member **25** that partitions a space interposed by the pair of boom side plates **21** in the vicinity of the center in the boom extension direction, a boom base end support member **26** that supports the boom base end portion on the upper swing body **6**, and an arm support plate **27** that supports the arm **30**.

(35) The boom side plate **21** includes a first cylinder base end side hole **21a** and a second cylinder base end side hole **21b** that are open in the vehicle width direction of the upper swing body **6**. As shown in FIG. **2**, the first cylinder base end side hole **21a** is disposed below the vicinity of the center in the boom extension direction and in a portion overlapping with a part of the first electric cylinder **100A** (in the vicinity of an upper end portion in FIG. **2**) when viewed from the vehicle width direction of the upper swing body **6**. The second cylinder base end side hole **21b** is disposed above the vicinity of the center in the boom extension direction and in a portion overlapping with a part of the second electric cylinder **100B** (in the vicinity of a lower end portion in FIG. **2**) when viewed from the vehicle width direction of the upper swing body **6**.

(36) As shown in FIG. **3**, the boom bottom plate **22** is provided at an edge portion on a side opposite to the upper swing body **6** in the boom plate width direction of the boom side plate **21**. The boom bottom plate **22** extends along the boom extension direction. The boom bottom plate **22** is curved between the first cylinder base end side hole **21a** and the second cylinder base end side hole **21b** in the vicinity of the center in the boom extension direction.

(37) The boom base end side connection plate **23** extends on the boom base end portion side in the vehicle width direction of the upper swing body **6**, to connect the pair of boom side plates **21** to each other. The boom base end side connection plate **23** extends to approach the boom base end portion as the distance from a connection portion with the boom bottom plate **22** increases in the boom plate width direction, and then is bent to extend toward the boom base end portion.

(38) The boom distal end side connection plate **24** extends on the boom distal end portion side in the vehicle width direction of the upper swing body **6**, to connect the pair of boom side plates **21** to each other. The boom distal end side connection plate **24** extends to approach the boom distal end portion as the distance from a connection portion with the boom bottom plate **22** increases in the boom plate width direction, and then is bent to extend toward the boom distal end portion. The boom distal end side connection plate **24** includes an opening portion **24a** that is open in the boom extension direction at a position adjacent to one boom side plate **21**.

(39) The boom partition member **25** extends in the vehicle width direction of the upper swing body **6** in the vicinity of the center in the boom extension direction, to connect the pair of boom side plates **21** to each other. The boom partition member **25** extends along the boom plate width direction. The boom partition member **25** is disposed between the first cylinder base end side hole **21a** and the second cylinder base end side hole **21b**. The boom partition member **25** is spaced apart from the boom bottom plate **22** in the boom plate width direction.

(40) The boom base end support member **26** is provided on the boom base end portion side. The boom base end support member **26** includes a first through-hole **26a** that is open in the vehicle width direction of the upper swing body **6**. A first pin **28** (see FIG. **2**) extending in the vehicle width direction of the upper swing body **6** is inserted into the first through-hole **26a**. The first pin **28** is inserted into the first through-hole **26a** of the boom base end support member **26** and the second hole **15b** of the bracket **15**, whereby the boom **20** is supported to be rotationally movable about a central axis **O1** of the first pin **28**.

(41) The arm support plate **27** is provided on the boom distal end portion side. The arm support portions **16** are provided on an outer surface of the boom side plate **21** to interpose the pair of boom side plates **21** from the outer side of the upper swing body **6** in the vehicle width direction. The arm

support plate **27** protrudes outward in the boom extension direction with respect to the boom side plate **21**. The arm support plate **27** includes a second through-hole **27a** that is open in the vehicle width direction of the upper swing body **6**. The second through-hole **27a** is provided in a portion of the arm support plate **27** that protrudes outward in the boom extension direction with respect to the boom side plate **21**. A second pin **29** (see FIG. 2) that extends in the vehicle width direction of the upper swing body **6** is inserted into the second through-hole **27a**.

(42) <Arm>

(43) In the posture of FIG. 1, the arm **30** extends downward and forward from a portion overlapping with a part of the second electric cylinder **100B** (in the vicinity of the upper end portion in FIG. 1) when viewed from the vehicle width direction of the upper swing body **6**. Hereinafter, when viewed from the vehicle width direction of the upper swing body **6**, a direction in which the arm **30** extends will be referred to as an “arm extension direction”, and a direction orthogonal to the arm extension direction will be referred to as an “arm plate width direction”. One end portion (end portion on the second electric cylinder **100B** side) of the arm **30** in the arm extension direction will be referred to as an “arm base end portion”. Another end portion (end portion on a side opposite to the arm base end portion) of the arm **30** in the arm extension direction will be referred to as an “arm distal end portion”. The dimension in the arm plate width direction gradually increases from the arm base end portion toward the vicinity of the boom connection portion in the arm extension direction, and then gradually decreases toward the arm distal end portion.

(44) As shown in FIG. 4, the arm **30** includes a pair of arm side plates **31** that are disposed to be spaced apart from each other in the vehicle width direction of the upper swing body **6**, an arm bottom plate **32** that extends in the vehicle width direction of the upper swing body **6** to connect the pair of arm side plates **31** to each other, an arm side connection plate **33** connected to the arm bottom plate **32**, an arm partition member **34** that partitions a space between the pair of arm side plates **31** in the vicinity of the arm base end portion, a boom distal end connection member **35** connected to the boom distal end portion, a bucket support member **36** that supports the bucket **40** (see FIG. 2), and a link support member **37** that supports one end portion of a first link member **41** (see FIG. 2).

(45) The arm side plate **31** includes a second cylinder distal end side hole **31a** and a third cylinder base end side hole **31b** that are open in the vehicle width direction of the upper swing body **6**. As shown in FIG. 2, the second cylinder distal end side hole **31a** is disposed in the vicinity of the arm base end portion and in a portion overlapping with a part of the second electric cylinder **100B** (in the vicinity of the upper end portion in FIG. 2) when viewed from the vehicle width direction of the upper swing body **6**. The third cylinder base end side hole **31b** is disposed on a side opposite to the portion overlapping with the boom distal end portion in the arm plate width direction and in a portion overlapping with a part of the third electric cylinder **100C** (in the vicinity of the upper end portion in FIG. 2) when viewed from the vehicle width direction of the upper swing body **6**.

(46) In a posture of FIG. 2, the arm bottom plate **32** is provided at an edge portion on the upper swing body **6** side (the boom **20** side) in the arm side plate **31** in the arm plate width direction. The arm bottom plate **32** extends along the arm extension direction. As shown in FIG. 4, the arm bottom plate **32** extends between the boom distal end connection member **35** and the bucket support member **36** in the arm extension direction.

(47) The arm side connection plate **33** extends on the arm distal end portion side in the vehicle width direction of the upper swing body **6**, to connect the pair of arm side plates **31** to each other. The arm side connection plate **33** extends to approach the arm distal end portion as the distance from a connection portion with the arm bottom plate **32** increases in the arm plate width direction, and then is bent to extend toward the arm distal end portion.

(48) The arm partition member **34** extends in the vicinity of the arm base end portion in the vehicle width direction of the upper swing body **6**, to connect the pair of arm side plates **31** to each other.

The arm partition member **34** is disposed between the second cylinder distal end side hole **31a** and the third cylinder base end side hole **31b**. The arm partition member **34** is disposed to be spaced apart from the boom distal end connection member **35**. When viewed from the vehicle width direction of the upper swing body **6**, the arm partition member **34** extends from the vicinity of the boom distal end connection member **35** toward the arm distal end portion side, and then is bent to extend to cross between the second cylinder distal end side hole **31a** and the third cylinder base end side hole **31b**.

(49) The boom distal end connection member **35** is formed in a tubular shape extending in the vehicle width direction of the upper swing body **6**. The boom distal end connection member **35** includes a boom connection hole **35a** that is open in the vehicle width direction of the upper swing body **6**. As shown in FIG. 2, the boom connection hole **35a** overlaps with the second through-hole **27a** of the arm support plate **27** when viewed from the vehicle width direction of the upper swing body **6**. The second pin **29** is inserted into the second through-hole **27a** of the arm support plate **27** and the boom connection hole **35a** of the boom distal end connection member **35**, whereby the arm **30** is supported to be rotationally movable about a central axis O2 (see FIG. 4) of the second pin **29**.

(50) As shown in FIG. 4, the bucket support member **36** is provided at the arm distal end portion. The bucket support member **36** is formed in a tubular shape extending in the vehicle width direction of the upper swing body **6**. The bucket support member **36** includes a third through-hole **36a** that is open in the vehicle width direction of the upper swing body **6**. A third pin **38** (see FIG. 2) extending in the vehicle width direction of the upper swing body **6** is inserted into the third through-hole **36a**.

(51) As shown in FIG. 4, the link support member **37** is disposed between the arm bottom plate **32** and the arm side connection plate **33**. The link support member **37** is disposed in the vicinity of the bucket support member **36**. The link support member **37** is formed in a tubular shape extending in the vehicle width direction of the upper swing body **6**. The link support member **37** protrudes outward in the vehicle width direction of the upper swing body **6** with respect to the pair of arm side plates **31**. The link support member **37** includes a first link connection hole **37a** that is open in the vehicle width direction of the upper swing body **6**.

(52) <Bucket>

(53) In the posture of FIG. 2, the bucket **40** is inclined from the arm distal end portion toward the vicinity of the center in the boom extension direction. The bucket **40** includes a bucket connection hole **40a** and a second link connection hole **40b** that are open in the vehicle width direction of the upper swing body **6**.

(54) The bucket connection hole **40a** overlaps with the third through-hole **36a** of the bucket support member **36** when viewed from the vehicle width direction of the upper swing body **6**. The third pin **38** is inserted into the third through-hole **36a** of the bucket support member **36** and the bucket connection hole **40a** of the bucket **40**, whereby the bucket **40** is supported to be rotationally movable about a central axis O3 (see FIG. 4) of the third pin **38**.

(55) In the posture of FIG. 2, the second link connection hole **40b** is disposed at a position lower and rearward of the bucket connection hole **40a**.

(56) <First Electric Cylinder>

(57) As shown in FIG. 2, the first electric cylinder **100A** is disposed on the boom base end portion side with respect to the boom partition member **25**. The first electric cylinder **100A** includes a first cylinder body **103A** configured to expand and contract along the boom extension direction, a first motor **101A** that is a driving source, and a first power transmission unit **102A** that transmits the driving force of the first motor **101A** to the first cylinder body **103A**.

(58) The first cylinder body **103A** and the first motor **101A** extend parallel to each other. A first end portion of the first cylinder body **103A** is connected to a pin **51** inserted into the first hole **15a** of the bracket **15**. The first electric cylinder **100A** is supported by the upper swing body **6** via the

bracket **15** to be rotationally movable about a central axis of the pin **51** extending in the width direction of the upper swing body **6**.

(59) A second end portion of the first cylinder body **103A** is connected to a pin **52** inserted into the first cylinder base end side hole **21a** of the boom **20**. The first electric cylinder **100A** is supported by the boom **20** to be rotationally movable about a central axis of the pin **52** extending in the width direction of the upper swing body **6**.

(60) The first motor **101A** is disposed on the second end portion side of the first cylinder body **103A**. The first motor **101A** is disposed on an inner side in the boom plate width direction with respect to the first cylinder body **103A**. The first motor **101A** operates the first cylinder body **103A** using a battery (not shown) provided in the upper swing body **6** as a power source. The first cylinder body **103A** expands and contracts in response to the driving of the first motor **101A**, whereby the boom **20** rotationally moves about the central axis O1 (see FIG. 3) of the first pin **28** with respect to the upper swing body **6**.

(61) A first wiring **61** extends from the first motor **101A**. The first wiring **61** extends along the boom base end side connection plate **23** and is led into the bracket **15**. The first wiring **61** is connected to a battery (not shown) through the bracket **15**.

(62) <Second Electric Cylinder>

(63) The second electric cylinder **100B** is disposed on the boom distal end portion side with respect to the boom partition member **25**. The second electric cylinder **100B** includes a second cylinder body **103B** configured to expand and contract along the boom extension direction, a second motor **101B** that is a driving source, and a second power transmission unit **102B** that transmits the driving force of the second motor **101B** to the second cylinder body **103B**.

(64) The second cylinder body **103B** and the second motor **101B** extend parallel to each other. A first end portion of the second cylinder body **103B** is connected to a pin **53** inserted into the second cylinder base end side hole **21b** of the boom **20**. The second electric cylinder **100B** is supported by the boom **20** to be rotationally movable about a central axis of the pin **53** extending in the width direction of the upper swing body **6** with respect to the boom **20**.

(65) A second end portion of the second cylinder body **103B** is connected to a pin **54** inserted into the second cylinder distal end side hole **31a** of the arm **30**. The second electric cylinder **100B** is supported by the arm **30** to be rotationally movable about a central axis of the pin **54** extending in the width direction of the upper swing body **6** with respect to the arm **30**.

(66) The second motor **101B** is disposed on the first end portion side of the second cylinder body **103B**. The second motor **101B** is disposed on an inner side in the boom plate width direction with respect to the second cylinder body **103B**. The second motor **101B** operates the second cylinder body **103B** using a battery (not shown) provided in the upper swing body **6** as a power source. The second cylinder body **103B** expands and contracts in response to the driving of the second motor **101B**, whereby the arm **30** rotationally moves about the central axis O2 (see FIG. 3) of the second pin **29** with respect to the boom **20**.

(67) A second wiring **62** extends from the second motor **101B**. The second wiring **62** extends toward the first motor **101A** and then extends along the boom base end side connection plate **23** together with the first wiring **61**, and is led into the bracket **15**. The second wiring **62** is connected to a battery (not shown) through the bracket **15**.

(68) <Third Electric Cylinder>

(69) The third electric cylinder **100C** is disposed on the arm distal end portion side with respect to the arm partition member **34**. The third electric cylinder **100C** includes a third cylinder body **103C** configured to expand and contract along the arm extension direction, a third motor **101C** that is a driving source, and a third power transmission unit **102C** that transmits the driving force of the third motor **101C** to the third cylinder body **103C**.

(70) The third cylinder body **103C** and the third motor **101C** extend parallel to each other. A first end portion of the third cylinder body **103C** is connected to a pin **55** inserted into the third cylinder

base end side hole **31b** of the arm **30**. The third electric cylinder **100C** is supported by the arm **30** to be rotationally movable about a central axis of the pin **55** extending in the width direction of the upper swing body **6** with respect to the arm **30**.

(71) A second end portion of the third cylinder body **103C** is connected to a first end portion of the first link member **41**. The first end portion of the first link member **41** includes a first link hole **41a** that is open in the width direction of the upper swing body **6**. The second end portion of the third cylinder body **103C** is connected to a pin **56** inserted into the first link hole **41a**. The third electric cylinder **100C** supports the first link member **41** to be rotationally movable about a central axis of the pin **56** extending in the width direction of the upper swing body **6** with respect to the first link member **41**.

(72) The second end portion of the first link member **41** includes a second link hole **41b** that is open in the width direction of the upper swing body **6**. A pin **57** is inserted into the second link hole **41b** together with the first link connection hole **37a** of the arm **30**. The first link member **41** is supported by the arm **30** to be rotationally movable about a central axis of the pin **57** extending in the width direction of the upper swing body **6** with respect to the arm **30**.

(73) The second end portion of the third cylinder body **103C** is connected to a first end portion of a second link member **42**. The first end portion of the second link member **42** includes a third link hole **42a** that is open in the width direction of the upper swing body **6**. The second end portion of the third cylinder body **103C** is connected to the pin **56** inserted into the third link hole **42a** together with the first link hole **41a**. The second link member **42** is provided to be rotationally movable about the central axis of the pin **56** extending in the width direction of the upper swing body **6** with respect to the second end portion of the third cylinder body **103C** and the first end portion of the first link member **41**.

(74) A second end portion of the second link member **42** includes a fourth link hole **42b** that penetrates in the width direction of the upper swing body **6**. A pin **58** is inserted into the fourth link hole **42b** together with the second link connection hole **40b** of the bucket **40**. The second link member **42** is provided to be rotationally movable about a central axis of the pin **58** extending in the width direction of the upper swing body **6** with respect to the bucket **40**.

(75) The third motor **101C** is disposed on the first end portion side of the third cylinder body **103C**. The third motor **101C** is disposed on an inner side in the arm plate width direction with respect to the third cylinder body **103C**. The third motor **101C** operates the third cylinder body **103C** using a battery (not shown) provided in the upper swing body **6** as a power source. The third cylinder body **103C** expands and contracts in response to the driving of the third motor **101C**, whereby the bucket **40** rotationally moves about the central axis O3 (see FIG. 4) of the third pin **38** with respect to the arm **30**.

(76) A third wiring **63** extends from the third motor **101C**. The third wiring **63** extends toward the boom **20** and then passes through the opening portion **24a** (see FIG. 3) of the boom distal end side connection plate **24**. Thereafter, the third wiring **63** extends toward the first motor **101A** and then extends along the boom base end side connection plate **23** together with the first wiring **61** and the second wiring **62**, and is led into the bracket **15**. The third wiring **63** is connected to a battery (not shown) through the bracket **15**.

(77) <Electric Cylinder>

(78) As shown in FIG. 1, the first electric cylinder **100A**, the second electric cylinder **100B**, and the third electric cylinder **100C** are the electric cylinder **100** common to each other. As shown in FIG. 5, the electric cylinder **100** includes a motor **101**, a power transmission unit **102**, and a cylinder body **103**.

(79) The motor **101** is a driving source of the electric cylinder **100**. For example, the motor **101** is a servo motor. As shown in FIG. 8, the motor **101** and the cylinder body **103** extend parallel to each other. The motor **101** and the cylinder body **103** are arranged apart from each other.

(80) The electric cylinder **100** includes an output shaft **105** that rotates in response to the driving of

the motor **101**. The output shaft **105** is provided coaxially with a central axis of the motor **101**. The output shaft **105** protrudes outward in the axial direction from an axial end surface **101f** of the motor **101**. Reference numeral C1 in the drawing indicates a motor axis along the central axis of the motor **101**.

(81) The power transmission unit **102** transmits the driving force of the motor **101** to a piston **182**. The power transmission unit **102** includes a planetary gear mechanism **110** that changes (for example, decelerates) the driving force of the output shaft **105**, and a transmission gear mechanism **120** that transmits the driving force changed by the planetary gear mechanism **110** to the piston **182**.

(82) <Planetary Gear Mechanism>

(83) As shown in FIG. 9, the planetary gear mechanism **110** includes a sun gear **111** (an exemplary example of a rotating body) coupled to the output shaft **105**, a plurality of planetary gears **112** disposed adjacent to the sun gear **111**, carriers **114** and **115** that rotatably support central shafts **113** (hereinafter, also referred to as “planetary shafts **113**”) of the plurality of planetary gears **112**, and a ring gear **116** surrounding the plurality of planetary gears **112**. The planetary gear mechanism **110** is covered with a tubular case **106** disposed adjacent to the axial end surface **101f** of the motor **101**.

(84) <Sun Gear>

(85) The sun gear **111** rotates in response to the rotation of the output shaft **105**. The sun gear **111** is formed in a tubular shape coaxial with the output shaft **105**. The length of the sun gear **111** in the axial direction is longer than a length of the output shaft **105** that protrudes from the axial end surface **101f** of the motor **101**. An axial base end portion (an end portion on the motor **101** side) of the sun gear **111** is spaced apart from the axial end surface **101f** of the motor **101**. The axial base end portion of the sun gear **111** has a larger diameter than an axial distal end portion (an end portion on a side opposite to the motor **101**) of the sun gear **111**.

(86) The sun gear **111** has a hollow portion **111a** that is open such that a lubricant can be accommodated. The hollow portion **111a** is open outward in the axial direction. The hollow portion **111a** is a space surrounded by the axial distal end portion of the output shaft **105** and an inner peripheral surface of the sun gear **111**. The hollow portion **111a** is provided between the axial distal end portion of the output shaft **105** and the axial distal end portion of the sun gear **111**.

(87) As shown in FIG. 14, the sun gear **111** includes a coupling tubular portion **111b** coupled to the output shaft **105**, and a gear side tubular portion **111c** coupled to an axial distal end portion of the coupling tubular portion **111b**. The coupling tubular portion **111b** is a portion of the sun gear **111** on the axial base end portion side. The coupling tubular portion **111b** is formed in a tubular shape coaxial with the output shaft **105**.

(88) A key groove **111d** into which a protrusion portion provided on an outer peripheral surface of the output shaft **105** can be fitted is formed on an inner peripheral surface of the coupling tubular portion **111b**. The key groove **111d** extends linearly in the axial direction of the output shaft **105**. In FIG. 14, the output shaft **105** is shown by a two-dot chain line. As shown in FIG. 15, the key groove **111d** is formed in a rectangular recess shape when viewed from the axial direction.

(89) As shown in FIG. 14, the coupling tubular portion **111b** includes a tapered portion **111e** that is inclined from an outer peripheral edge of the coupling tubular portion **111b** toward the axial base end portion of the gear side tubular portion **111c**. An outer peripheral surface of the tapered portion **111e** is inclined to be located on an inner side in the radial direction as the outer peripheral surface goes from the outer peripheral edge of the coupling tubular portion **111b** toward the axial end portion of the gear side tubular portion **111c**.

(90) The gear side tubular portion **111c** is a portion of the sun gear **111** having outer teeth. The gear side tubular portion **111c** is a portion of the sun gear **111** on the axial distal end portion side. The gear side tubular portion **111c** extends outward in the axial direction from the axial distal end portion of the tapered portion **111e**. The gear side tubular portion **111c** is formed in a tubular shape coaxial with the coupling tubular portion **111b**. The gear side tubular portion **111c** and the coupling

tubular portion **111b** are integrally formed of the same member. As shown in FIG. **10**, a tooth surface of the sun gear **111** (an outer tooth surface of the sun gear **111**) is in contact with a tooth surface of the planetary gear **112** (an outer tooth surface of the planetary gear **112**).

(91) <Planetary Gear>

(92) The planetary gear **112** rotates in response to the rotation of the sun gear **111**. As shown in FIG. **10**, a plurality (for example, three in the present embodiment) of planetary gears **112** are disposed to be spaced apart from each other at equal intervals along a circumferential direction of the sun gear **111**. The outer teeth provided on an outer periphery of the planetary gear **112** mesh with the outer teeth provided on an outer periphery of the sun gear **111**. The planetary gear **112** rotates and revolves around the sun gear **111** by meshing with the sun gear **111**. The planetary gear **112** can rotate about the planetary shaft **113** that extends parallel to the output shaft **105**.

(93) <Carrier>

(94) As shown in FIG. **9**, the carriers **114** and **115** are provided coaxially with the output shaft **105**. The carriers **114** and **115** support both axial end portions of the planetary shaft **113**. The carriers **114** and **115** include guide grooves **143a** and **151a** that extend from a position facing the outer periphery of the sun gear **111** toward the planetary shaft **113** and are recessed to allow the lubricant to flow.

(95) The carriers **114** and **115** are a first carrier **114** disposed on the axial distal end portion side of the output shaft **105** and a second carrier **115** disposed on the axial central side of the output shaft **105**. As shown in FIG. **10**, the first carrier **114** and the second carrier **115** are coupled to each other by a plurality (for example, six in the present embodiment) of bolts **145**.

(96) <First Carrier>

(97) As shown in FIG. **11**, the first carrier **114** includes a first shaft hole **114a** that is open such that the planetary shaft **113** can be inserted, and a first bolt hole **114b** that is open such that the bolt **145** (see FIG. **10**) can be inserted. The first carrier **114** includes an annular first carrier base portion **140**, a tubular carrier distal end tubular body **141** (see FIG. **9**) that protrudes outward in the axial direction from the first carrier base portion **140**, a plurality of carrier wall portions **142** that extend inward in the axial direction (in a direction opposite to a direction in which the carrier distal end tubular body **141** protrudes) from the first carrier base portion **140**, and a first groove forming portion **143** that is provided between two carrier wall portions **142** adjacent to each other in the circumferential direction. The first carrier base portion **140**, the carrier distal end tubular body **141**, the carrier wall portion **142**, and the first groove forming portion **143** are integrally formed of the same member.

(98) As shown in FIG. **9**, an outer diameter of the first carrier base portion **140** is larger than an outer diameter of the carrier distal end tubular body **141**. An outer peripheral edge of the first carrier base portion **140** is spaced apart from an inner peripheral surface of the case **106** toward the inner side in the radial direction.

(99) As shown in FIG. **11**, the first carrier base portion **140** includes an annular groove **140a** disposed at a position facing the axial distal end portion of the output shaft **105**, and a plurality of relay grooves **140b** connected to the annular groove **140a**. The annular groove **140a** is formed in an annular shape along an inner periphery of the first carrier base portion **140**. The relay groove **140b** is curved outward in the radial direction from an outer peripheral edge of the annular groove **140a**.

(100) The plurality (for example, three in the present embodiment) of carrier wall portions **142** are disposed to be spaced apart from each other at equal intervals along a circumferential direction of the first carrier base portion **140**. The carrier wall portion **142** is provided between the outer peripheral edge of the annular groove **140a** and the outer peripheral edge of the first carrier base portion **140**. The carrier wall portion **142** has an outer shape that bulges in the circumferential direction of the first carrier base portion **140** toward the radial outer side of the first carrier base portion **140** when viewed from the axial direction. A side surface of the carrier wall portion **142** in the circumferential direction of the first carrier base portion **140** is curved in an arc shape along the

outer shape of the first groove forming portion **143**. Two first bolt holes **114b** are provided in each carrier wall portion **142**.

(101) The plurality (for example, three in the present embodiment) of first groove forming portions **143** are disposed to be spaced apart from each other at equal intervals along the circumferential direction of the first carrier base portion **140**. The first groove forming portion **143** is spaced apart from side surfaces of the two carrier wall portions **142** adjacent to each other in the circumferential direction at equal intervals. The first groove forming portion **143** is provided at a position overlapping with the planetary gear **112** when viewed from the axial direction.

(102) The first groove forming portion **143** includes a first guide groove **143a** that is recessed to allow the lubricant to flow. The first guide groove **143a** is formed on an imaginary line connecting an axial center of the carrier distal end tubular body **141** and an axial center of the first shaft hole **114a** when viewed from the axial direction. As shown in FIG. 9, the first guide groove **143a** extends from a position facing the axial distal end portion of the sun gear **111** toward the first shaft hole **114a**. As shown in FIG. 11, the first guide groove **143a** is disposed between the first shaft hole **114a** and the relay groove **140b**. The depth of the first guide groove **143a** gradually increases from the first shaft hole **114a** side toward the relay groove **140b**.

(103) The first groove forming portion **143** includes a first reception surface **143b** that receives an axial outer end surface of the planetary gear **112**. The first reception surface **143b** is formed in a C-shape that is open at a portion of the first guide groove **143a** when viewed from the axial direction. The first reception surface **143b** is configured to be in contact with the axial outer end surface of the planetary gear **112** in a portion other than the first guide groove **143a**. An inner peripheral edge of the first reception surface **143b** is spaced apart from an outer peripheral edge of the first shaft hole **114a** toward the outer side in the radial direction.

(104) <Second Carrier>

(105) As shown in FIG. 12, the second carrier **115** includes a second shaft hole **115a** that is open such that the planetary shaft **113** can be inserted, and a second bolt hole **115b** that is open such that the bolt **145** (see FIG. 10) can be inserted. The second carrier **115** includes an annular second carrier base portion **150** and a second groove forming portion **151** that is provided at a position facing the first groove forming portion **143** (see FIG. 11) in the axial direction. The second carrier base portion **150** and the second groove forming portion **151** are integrally formed of the same material.

(106) As shown in FIG. 9, an outer diameter of the second carrier base portion **150** is substantially the same as the outer diameter of the first carrier base portion **140**. An outer peripheral edge of the second carrier base portion **150** is spaced apart from the inner peripheral surface of the case **106** toward the inner side in the radial direction.

(107) The second carrier base portion **150** has an opening portion **150a** that is open such that the output shaft **105** can be inserted. The second carrier base portion **150** includes a wall reception portion **150b** that receives the carrier wall portion **142** of the first carrier **114**.

(108) The wall reception portion **150b** is provided at a position overlapping with the carrier wall portion **142** when viewed from the axial direction. As shown in FIG. 12, a plurality (for example, three in the present embodiment) of wall reception portions **150b** are provided to correspond to the carrier wall portions **142**. Two second bolt holes **115b** are provided in each wall reception portion **150b**.

(109) The plurality (for example, three in the present embodiment) of second groove forming portions **151** are disposed to be spaced apart from each other at equal intervals along a circumferential direction of the second carrier base portion **150**. The second groove forming portion **151** is provided at a position overlapping with the planetary gear **112** when viewed from the axial direction.

(110) The second groove forming portion **151** includes a second guide groove **151a** that is recessed to allow the lubricant to flow. The second guide groove **151a** is formed on an imaginary line

connecting a center of the opening portion **150a** and an axial center of the second shaft hole **115a** when viewed from the axial direction. As shown in FIG. **9**, the second guide groove **151a** extends from a position facing the outer periphery of the sun gear **111** toward the second shaft hole **115a**. As shown in FIG. **12**, the second guide groove **151a** is disposed between the second shaft hole **115a** and the opening portion **150a**. A depth of the second guide groove **151a** gradually increases from the second shaft hole **115a** side toward the opening portion **150a**.

(111) The second groove forming portion **151** includes a second reception surface **151b** that receives an axial inner end surface of the planetary gear **112**. The second reception surface **151b** is formed in a C-shape that is open at a portion of the second guide groove **151a** when viewed from the axial direction. The second reception surface **151b** is configured to be in contact with the axial inner end surface of the planetary gear **112** in a portion other than the second guide groove **151a**. An inner peripheral edge of the second reception surface **151b** is spaced apart from an outer peripheral edge of the second shaft hole **115a** toward the outer side in the radial direction.

(112) <Ring Gear>

(113) As shown in FIG. **10**, inner teeth provided on an inner periphery of the ring gear **116** mesh with the outer teeth provided on the outer periphery of each planetary gear **112**. A plurality of gear side recess portions **116a** into which the rotation stop pins **117** enter are provided in an outer peripheral surface of the ring gear **116**. The plurality (for example, four in the present embodiment) of gear side recess portions **116a** are spaced apart from each other at equal intervals in the circumferential direction.

(114) A plurality of case side recess portions **106a** into which the rotation stop pins **117** enter are provided in an inner peripheral surface of the case **106**. The plurality (for example, four in the present embodiment) of case side recess portions **106a** are spaced apart from each other at equal intervals in the circumferential direction. For example, by inserting the rotation stop pin **117** into each of the recess portions **106a** and **116a** in a state in which the circumferential positions of the gear side recess portion **116a** and the case side recess portion **106a** are aligned with each other, the rotation of the ring gear **116** can be stopped (the ring gear **116** is restricted from moving in the circumferential direction with respect to the case **106**).

(115) <Holder>

(116) As shown in FIG. **9**, the electric cylinder **100** includes a holder **118** disposed adjacent to the axial end surface **101f** of the motor **101**. The holder **118** is disposed between the axial end surface **101f** of the motor **101** and the carriers **114** and **115**. The holder **118** is formed to surround the space between the axial end surface **101f** of the motor **101** and the second carrier **115** from the outer side in the radial direction.

(117) As shown in FIG. **17**, the holder **118** includes a fluid accommodation portion **201** that can accommodate a magnetic fluid **200** between the holder **118** and the outer periphery of the sun gear **111**. For example, the magnetic fluid **200** is a liquid in which magnetic fine particles, such as magnetite, are dispersed in a base liquid by a surfactant. The thermal conductivity of the magnetic fluid **200** is higher than the thermal conductivity of air. It should be noted that the aspect of the magnetic fluid **200** can be changed depending on the required specifications.

(118) The holder **118** is formed in a tubular shape. The holder **118** includes a through-hole **118a** that is open in the axial direction of the motor **101** with a gap from the outer peripheral edge (radial outer end edge) of the coupling tubular portion **111b** of the sun gear **111**. The gap of the through-hole **118a** is formed to have a size through which the lubricant can flow. The holder **118** includes an inner recess portion **118b** that is open inward in the axial direction at a position facing the axial end surface **101f** of the motor **101**, an outer recess portion **118c** that is open outward in the axial direction such that the second carrier **115** can be accommodated, and an accommodation recess portion **119** that is open outward in the axial direction at a position facing the fluid accommodation portion **201**. In FIG. **17**, the motor **101** and the output shaft **105** are indicated by two-point chain lines.

(119) The through-hole **118a** allows the radial central portions (portions on the sun gear **111** side) of the inner recess portion **118b** and the accommodation recess portion **119** to communicate with each other in the axial direction. An axial inner end portion of the through-hole **118a** is connected to the radial central portion of the inner recess portion **118b**. An axial outer end portion of the through-hole **118a** is connected to the radial central portion of the accommodation recess portion **119**. An inner diameter of the inner recess portion **118b** is larger than an inner diameter of the outer recess portion **118c**. An inner diameter of the accommodation recess portion **119** is smaller than the inner diameter of the outer recess portion **118c**. An inner peripheral edge of the outer recess portion **118c** is spaced apart from an outer peripheral edge of the second carrier **115** toward the outer side in the radial direction. An inner peripheral surface of the outer recess portion **118c** is disposed on the axial outer side of the motor **101** with respect to the fluid accommodation portion **201**. The inner peripheral surface of the outer recess portion **118c** is disposed on the radial outer side of the sun gear **111** with respect to the fluid accommodation portion **201**. An inner peripheral surface of the accommodation recess portion **119** is disposed on the radial inner side of the sun gear **111** with respect to the inner peripheral surface of the outer recess portion **118c**.

(120) As shown in FIG. **13**, the accommodation recess portion **119** has a support surface **119a** formed in an annular shape coaxial with the sun gear **111**. As shown in FIG. **17**, the support surface **119a** extends in the radial direction of the sun gear **111**. The support surface **119a** is formed parallel to an axial inner surface of the inner recess portion **118b**.

(121) The holder **118** includes a holder side groove **118d** that extends from the inner peripheral edge of the outer recess portion **118c** toward an axial inner end portion of the ring gear **116** and is recessed to allow the lubricant to flow. The depth of the holder side groove **118d** gradually increases from the inner peripheral edge side of the outer recess portion **118c** toward the axial inner end portion of the ring gear **116**.

(122) <Magnetic Pole Piece>

(123) The electric cylinder **100** includes a pair of magnetic pole pieces **211** and **212** provided on an inner peripheral surface of the holder **118**. The pair of magnetic pole pieces **211** and **212** are provided on the inner peripheral surface of the accommodation recess portion **119**. As shown in FIG. **13**, the pair of magnetic pole pieces **211** and **212** are formed in an annular shape coaxial with the sun gear **111**. The pair of magnetic pole pieces **211** and **212** are a first magnetic pole piece **211** disposed between the support surface **119a** of the holder **118** and a magnet **210**, and a second magnetic pole piece **212** disposed adjacent to a surface of the magnet **210** opposite to the first magnetic pole piece **211**. The first magnetic pole piece **211** and the second magnetic pole piece **212** are formed in a ring shape having the same size as each other.

(124) As shown in FIG. **17**, for example, it is preferable that outer peripheral surfaces of the magnetic pole pieces **211** and **212** are in contact with the inner peripheral surface of the accommodation recess portion **119**. As a result, the radial misregistration of the magnetic pole pieces **211** and **212** with respect to the inner peripheral surface of the accommodation recess portion **119** can be suppressed.

(125) <Magnet>

(126) The electric cylinder **100** includes the magnet **210** disposed between the pair of magnetic pole pieces **211** and **212**. The magnet **210** holds the magnetic fluid **200** in the fluid accommodation portion **201** by using the action of the magnetic field. As shown in FIG. **13**, the magnet **210** is formed in an annular shape coaxial with the sun gear **111**. The magnet **210** has a ring shape magnetized in the axial direction. The magnet **210** is formed in a ring shape having a larger inner diameter larger than the magnetic pole pieces **211** and **212**. The magnet **210** has the same outer diameter as the magnetic pole pieces **211** and **212**. As shown in FIG. **17**, an inner peripheral surface of the magnet **210** is disposed on the radial outer side of the sun gear **111** with respect to the inner peripheral surfaces of the magnetic pole pieces **211** and **212**.

(127) For example, it is preferable that the magnet **210** is in contact with the inner peripheral

surface of the accommodation recess portion **119**. As a result, it is possible to suppress the radial misregistration of the magnet **210** with respect to the inner peripheral surface of the accommodation recess portion **119**.

(128) The magnet **210** holds the magnetic fluid **200** in the fluid accommodation portion **201** in cooperation with the first magnetic pole piece **211** and the second magnetic pole piece **212**. The magnet **210**, the first magnetic pole piece **211**, and the second magnetic pole piece **212** constitute a fluid holding structure **209** that holds the magnetic fluid **200** in the fluid accommodation portion **201**. In the present embodiment, one fluid holding structure **209** is provided on the inner peripheral surface of the accommodation recess portion **119** of the holder **118**.

(129) <Retaining Ring>

(130) The electric cylinder **100** includes a retaining ring **215** that stops the second magnetic pole piece **212** from coming off outward in the axial direction of the motor **101**. For example, the retaining ring **215** is a snap ring. The retaining ring **215** is disposed adjacent to a surface of the second magnetic pole piece **212** opposite to the magnet **210**. As shown in FIG. **16**, the retaining ring **215** is formed in a C-shape when viewed from the axial direction. The retaining ring **215** is formed in an arc shape that overlaps with an outer peripheral portion of the second magnetic pole piece **212** when viewed from the axial direction.

(131) As shown in FIG. **17**, the holder **118** includes a fitting groove **119b** that is recessed to allow the retaining ring **215** to be fitted. The fitting groove **119b** is disposed in the vicinity of the axial outer end on the inner peripheral surface of the accommodation recess portion **119**. The fitting groove **119b** is recessed outward in the radial direction over the entire circumference of the inner peripheral surface of the accommodation recess portion **119**. The fitting groove **119b** is formed in an annular shape along the inner periphery of the accommodation recess portion **119**. The fitting groove **119b** is formed in an annular shape along an outer periphery of the retaining ring **215**. The first magnetic pole piece **211**, the magnet **210**, and the second magnetic pole piece **212** are disposed and arranged in this order in the axial direction between the support surface **119a** of the holder **118** and the retaining ring **215**.

(132) For example, in a state in which the retaining ring **215** is fitted into the fitting groove **119b**, it is preferable that the surfaces of the support surface **119a** of the accommodation recess portion **119**, the first magnetic pole piece **211**, the magnet **210**, the second magnetic pole piece **212**, and the retaining ring **215** are in contact with each other, the surfaces being adjacent to each other in the axial direction. As a result, the misregistration of the first magnetic pole piece **211**, the magnet **210**, and the second magnetic pole piece **212** in the axial direction with respect to the support surface **119a** of the accommodation recess portion **119** can be suppressed.

(133) <Fluid Accommodation Portion>

(134) The fluid accommodation portion **201** is provided to surround the outer periphery (the outer periphery of the coupling tubular portion **111b**) of the sun gear **111** on the axial base end portion side in the circumferential direction. The fluid accommodation portion **201** is partitioned by the outer peripheral surface (an outer peripheral surface of the coupling tubular portion **111b**) of the sun gear **111** on the axial base end portion, the inner peripheral surfaces of the pair of magnetic pole pieces **211** and **212**, and the inner peripheral surface of the magnet **210**. The fluid accommodation portion **201** includes gaps **201a** and **201b** that allow the magnetic fluid **200** to be accommodated between the outer peripheral surface of the sun gear **111** and the inner peripheral surfaces of the magnetic pole pieces **211** and **212**, and a space **201c** that allows the magnetic fluid **200** to be accommodated between the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the magnet **210**.

(135) Hereinafter, the gap **201a** between the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the first magnetic pole piece **211** will also be referred to as a “first gap **201a**”, and the gap **201b** between the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the second magnetic pole piece **212** will also be referred to as a “second gap

201b". The first gap **201a** and the second gap **201b** communicate with each other through the space **201c**. The magnetic fluid **200** is accommodated in the first gap **201a**, the second gap **201b**, and the space **201c**.

(136) For example, the sun gear **111** may be formed of a magnetic material. For example, at least a portion (the coupling tubular portion **111b**) of the sun gear **111** facing the fluid accommodation portion **201** may be formed of a magnetic material. In this configuration, due to the action of the magnetic field caused by the magnet **210**, the magnetic pole pieces **211** and **212**, and the sun gear **111**, the magnetic fluid **200** can be held in the fluid accommodation portion **201**.

(137) <Capturing Portion>

(138) The electric cylinder **100** includes a capturing portion **216** that can capture the lubricant attempting to flow from the axial distal end portion side of the sun gear **111** toward the fluid accommodation portion **201**. The capturing portion **216** protrudes from the outer peripheral surface of the sun gear **111** toward the inner peripheral surface of the outer recess portion **118c**. As shown in FIG. **16**, the capturing portion **216** is formed in an annular shape coaxial with the sun gear **111**. For example, the capturing portion **216** is an annular member that can be attached to the outer peripheral surface of the sun gear **111** by press-fitting.

(139) As shown in FIG. **14**, the capturing portion **216** includes a first annular portion **216a** that has a ring shape along the outer periphery of the coupling tubular portion **111b** of the sun gear **111**, and a second annular portion **216b** that has a ring shape coupled to the first annular portion **216a** and extending along the outer periphery of the tapered portion **111e**. The first annular portion **216a** and the second annular portion **216b** are integrally formed of the same member.

(140) An inner peripheral surface of the first annular portion **216a** is in contact with the outer peripheral surface of the coupling tubular portion **111b**. An inner peripheral surface of the second annular portion **216b** is in contact with the outer peripheral surface of the tapered portion **111e**. The inner peripheral surface of the second annular portion **216b** is inclined from the inner peripheral edge of the first annular portion **216a** along the outer peripheral surface of the tapered portion **111e**. As shown in FIG. **17**, an axial distal end surface of the second annular portion **216b** extends in the radial direction of the sun gear **111**.

(141) In a cross-sectional view of FIG. **17**, a cross-sectional shape of the capturing portion **216** on the radial distal end side is a rectangular protruding shape. It should be noted that the cross-sectional shape of the capturing portion **216** on the radial distal end side may be a shape other than the rectangular protruding shape, such as an arc shape, a trapezoidal shape, or a triangular shape. For example, the cross-sectional shape of the capturing portion **216** on the radial distal end side can be changed depending on the required specifications.

(142) <Transmission Gear Mechanism>

(143) As shown in FIG. **8**, the transmission gear mechanism **120** includes a transfer gear **121** that transmits the rotational force of the carriers **114** and **115** to the piston **182**, a transfer shaft **122** that extends outward in the axial direction from a position facing an axial outer end of the sun gear **111**, an idler gear **123** that is disposed adjacent to the transfer gear **121**, and a driven gear **124** that is disposed on a side opposite to the transfer gear **121** across the idler gear **123**. The transmission gear mechanism **120** is covered with a cover unit **160** disposed adjacent to the case **106**.

(144) As shown in FIG. **9**, the transfer gear **121** is provided coaxially with the output shaft **105**. The transfer gear **121** is formed in a tubular shape that is open such that the transfer shaft **122** can be inserted. The transfer gear **121** includes a tubular gear body **121a** having outer teeth that mesh with the idler gear **123**, an inner tubular body **121b** that protrudes inward in the axial direction from the gear body **121a**, and an outer tubular body **121c** that protrudes outward in the axial direction from the gear body **121a**. The gear body **121a**, the inner tubular body **121b**, and the outer tubular body **121c** are integrally formed of the same member. The transfer gear **121** is rotatably supported by the cover unit **160** about the motor axis **C1** by an inner bearing **130** provided on the outer periphery of the inner tubular body **121b** and an outer bearing **131** provided on the outer periphery of the outer

tubular body **121c**.

(145) The transfer shaft **122** is provided coaxially with the output shaft **105**. The carrier distal end tubular body **141** is coupled to the transfer shaft **122** on the one end portion side in the axial direction via a spline. The inner teeth having the tooth surfaces parallel to the axial direction of the carrier distal end tubular body **141** are provided on an inner periphery of the carrier distal end tubular body **141**. The outer teeth having the tooth surfaces parallel to the axial direction of the transfer shaft **122** and meshing with the inner teeth of the carrier distal end tubular body **141** are provided on the outer periphery of the transfer shaft **122** on the one end portion side in the axial direction. A gap through which the lubricant can flow is formed between the outer teeth of the transfer shaft **122** on the one end portion side in the axial direction and the inner teeth of the carrier distal end tubular body **141**.

(146) The gear body **121a** of the transfer gear **121** is coupled to the transfer shaft **122** on the other end portion side in the axial direction via a spline. The inner teeth having the tooth surfaces parallel to the axial direction of the transfer gear **121** are provided on an inner periphery of the gear body **121a**. The outer teeth having the tooth surfaces parallel to the axial direction of the transfer shaft **122** and meshing with the inner teeth of the gear body **121a** are provided on the outer periphery of the transfer shaft **122** on the other end portion side in the axial direction. A gap through which the lubricant can flow is formed between the outer teeth of the transfer shaft **122** on the other end portion side in the axial direction and the inner teeth of the gear body **121a**.

(147) An axial inner end portion of the inner tubular body **121b** is connected to a distal end portion of the carrier distal end tubular body **141** via an O-ring **132**. The bearing **133** is provided between an inner periphery of the inner tubular body **121b** and an outer periphery of an axial central portion of the transfer shaft **122**. For example, the bearing **133** is composed of a pair of semi-circular arc-shaped rings (so-called half-split rings). A gap through which the lubricant can flow is formed between the axial central portion of the transfer shaft **122** and the bearing **133**.

(148) A cover member **135** is attachably and detachably attached to the outer tubular body **121c**. The cover member **135** includes a supply hole **135a** that is open such that the lubricant can be supplied from the outside to the other end portion side in the axial direction of the transfer shaft **122**. The supply hole **135a** is formed on the motor axis C1. A gap through which the lubricant can flow is formed between an axial outer end portion of the transfer shaft **122** and the cover member **135**.

(149) A grease nipple **136** that can be opened and closed such that the lubricant can be supplied from the outside to the supply hole **135a** is provided in the cover member **135**. The grease nipple **136** is provided on the motor axis C1. The grease nipple **136** extends outward in the axial direction with respect to the cover member **135**. The grease nipple **136** includes an injection port (not shown) of the lubricant that communicates with the supply hole **135a**. The grease nipple **136** has a check valve in which a ball is pressed by a spring from the inside with respect to the injection port. For example, the grease nipple **136** can be opened by applying pressure to the grease nipple **136** by connecting a grease gun or the like, and the lubricant can be supplied to the supply hole **135a** through the injection port.

(150) The idler gear **123** rotates in response to the rotation of the transfer gear **121**. The idler gear can rotate about an idler shaft **123a** extending parallel to the transfer shaft **122**. The idler gear **123** is formed in a tubular shape that is open such that the idler shaft **123a** can be inserted. A bearing **123b** is provided between an inner periphery of the idler gear **123** and an outer periphery of the idler shaft **123a**.

(151) As shown in FIG. 8, the driven gear **124** is disposed adjacent to the idler gear **123**. The driven gear **124** rotates in response to the rotation of the idler gear **123**. The driven gear **124** is provided coaxially with a cylinder shaft **180** accommodated inside the cylinder body **103**. Reference numeral C2 in the drawing indicates a cylinder axis along the cylinder shaft **180**.

(152) The driven gear **124** is formed in a tubular shape that is open such that a first end portion of

the cylinder shaft **180** can be inserted. The driven gear **124** includes a tubular gear body **124a** having outer teeth that mesh with the idler gear **123**, an inner tubular body **124b** that protrudes inward in the axial direction from the gear body **124a**, and an outer tubular body **124c** that protrudes outward in the axial direction from the gear body **124a**. The gear body **124a**, the inner tubular body **124b**, and the outer tubular body **124c** are integrally formed of the same member.

(153) The driven gear **124** is rotatably supported by the cover unit **160** about the cylinder axis C2 by an inner bearing **155** provided on the outer periphery of the inner tubular body **124b** and an outer bearing **156** provided on the outer periphery of the outer tubular body **124c**.

(154) In the drawing, reference numeral **137** indicates a cover member provided to be attachable to and detachable from the outer tubular body **124c**, and reference numeral **138** indicates a grease nipple that is provided on the cover member **137** and that can be opened and closed such that the lubricant can be supplied from the outside to the supply hole of the cover member **137**.

(155) <Cover Unit>

(156) The cover unit **160** includes a first cover **161** that covers the transfer gear **121** from the outer side in the axial direction, a second cover **162** that covers the driven gear **124** from the outer side in the axial direction, and a third cover **163** that covers the transfer gear **121**, the idler gear **123**, and the driven gear **124** from the radial outer side each gear.

(157) As shown in FIG. 7, the first cover **161** has a rectangular shape when viewed from the axial direction. As shown in FIG. 9, the first cover **161** includes a first supply opening **161a** that is open such that the lubricant can be supplied from the outside to the other end portion side in the axial direction of the transfer shaft **122**. The first supply opening **161a** is formed on the motor axis C1. A gap through which the lubricant can flow is formed between an axial outer end portion of the transfer gear **121** and the first cover **161**. A first lid member **165** is attachably and detachably attached to the first cover **161** such that the first supply opening **161a** can be opened and closed.

(158) As shown in FIG. 8, the second cover **162** includes a second supply opening **162a** that is open such that the lubricant can be supplied from the outside to the first end portion side of the cylinder shaft **180**. The second supply opening **162a** is formed on the cylinder axis C2. A gap through which the lubricant can flow is formed between an axial outer end portion of the driven gear **124** and the second cover **162**. A second lid member **166** is attachably and detachably attached to the second cover **162** such that the second supply opening **162a** can be opened and closed.

(159) As shown in FIG. 7, the second cover **162** includes an idler cover portion **162b** provided at a position overlapping with the idler gear **123** and a driven cover portion **162c** provided at a position overlapping with the driven gear **124**, when viewed from the axial direction. The idler cover portion **162b** and the driven cover portion **162c** are integrally formed of the same material. As shown in FIG. 9, the idler cover portion **162b** fixes the idler shaft **123a** via a bolt **170**.

(160) As shown in FIG. 5, the third cover **163** includes a case side cover portion **163a** provided between the case **106** and the first cover **161**, and a screw side cover portion **163b** provided between the cylinder body **103** and the second cover **162**.

(161) As shown in FIG. 9, the case side cover portion **163a** is opened coaxially with the motor axis C1. In the drawing, reference numeral **167** indicates an inner spacer provided between an inner peripheral surface of an axial inner portion of the case side cover portion **163a** and the inner bearing **130**, and reference numeral **168** indicates an outer spacer provided between an inner peripheral surface of an axial outer portion of the case side cover portion **163a** and the outer bearing **131**.

(162) As shown in FIG. 5, the first cover **161** is jointly fastened to the case **106** via the case side cover portion **163a** by using a plurality (for example, four in the present embodiment) of bolts **171**. An axial inner end portion of the case side cover portion **163a** is coupled to an axial outer end portion of the case **106** via the joint fastening using the bolt **171**.

(163) The driven cover portion **162c** is fixed to the screw side cover portion **163b** via a plurality (for example, eight in the present embodiment) of bolts **172**. The screw side cover portion **163b** is

fixed to the cylinder body **103** via a plurality (for example, four in the present embodiment) of bolts **173**.

(164) <Cylinder Body>

(165) As shown in FIG. **8**, the cylinder body **103** includes the cylinder shaft **180**, a nut **181** screwed onto a screw shaft **180a** of the cylinder shaft **180**, the piston **182** provided on an outer periphery of the nut **181**, a tubular piston rod **183** coupled to the piston **182**, a joint member **184** provided at a distal end portion of the piston rod **183**, a tubular cylinder tube **185** that accommodates the piston rod **183**, a holding member **186** provided between a first end portion of the cylinder tube **185** and the screw side cover portion **163b**, and a rod cover **187** provided at a second end portion of the cylinder tube **185**.

(166) A ball (not shown) intervenes between the screw shaft **180a** and the nut **181**. The screw shaft **180a** and the nut **181** constitute a ball screw that converts rotational movement of the motor **101** into linear movement. The nut **181** is coupled to the piston **182** via a plurality of bolts. The piston **182** is configured to move integrally with the nut **181** on the screw shaft **180a**. The piston rod **183** is configured to move integrally with the piston **182** along the cylinder axis C2.

(167) As shown in FIG. **6**, the joint member **184** protrudes outward from an outer peripheral edge of the rod cover **187**. As shown in FIG. **8**, the joint member **184** has a connection hole **184a** that is open in a direction orthogonal to the cylinder axis C2. A bearing **188** is provided between an inner peripheral surface of the cylinder tube **185** and an outer peripheral surface of the piston **182**.

(168) As shown in FIG. **5**, the holding member **186** includes a tubular holding portion body **190** and a trunnion portion **191** that protrudes outward in the radial direction from the holding portion body **190**.

(169) As shown in FIG. **8**, the holding portion body **190** is opened coaxially with the cylinder axis C2. A plurality of bearings **189** are provided between an inner peripheral surface of the holding portion body **190** and the screw shaft **180a**. As shown in FIG. **5**, the trunnion portion **191** includes a connection hole **191a** that is open in a direction orthogonal to the cylinder axis C2. The connection hole **191a** of the trunnion portion **191** is open parallel to the connection hole **184a** of the joint member **184**.

(170) <Operation of Electric Cylinder>

(171) Hereinafter, an exemplary example of an operation of the electric cylinder **100** will be described.

(172) As shown in FIG. **8**, the driving force from the motor **101** is decelerated through the power transmission unit **102** and is transmitted to the cylinder shaft **180**. Specifically, the driving force from the motor **101** is the rotational force about the motor axis C1, and is decelerated by the output shaft **105**, the sun gear **111**, the plurality of planetary gears **112**, and the carriers **114** and **115**. The rotational force decelerated by the carriers **114** and **115** is transmitted to the transfer gear **121** through the transfer shaft **122**. The rotational force transmitted to the transfer gear **121** is transmitted to the cylinder shaft **180** through the idler gear **123** and the driven gear **124**.

(173) For example, in a case in which the motor **101** rotates in a forward direction, the cylinder shaft **180** rotates in one direction about the cylinder axis C2. The nut **181** screwed onto the screw shaft **180a** of the cylinder shaft **180** moves on the cylinder axis C2 in a direction of an arrow M1 in response to the rotation of the cylinder shaft **180** in the one direction. The piston **182**, the piston rod **183**, and the joint member **184** integrally move in the direction of the arrow M1 in response to the movement of the nut **181** in the direction of the arrow M1. As a result, the cylinder body **103** expands.

(174) On the other hand, in a case in which the motor **101** rotates in a reverse direction from a state in which the cylinder body **103** expands, the cylinder shaft **180** rotates in the other direction about the cylinder axis C2. The nut **181** screwed onto the screw shaft **180a** of the cylinder shaft **180** moves on the cylinder axis C2 in a direction opposite to the direction of the arrow M1 in response to the rotation of the cylinder shaft **180** in the other direction. The piston **182**, the piston rod **183**,

and the joint member **184** integrally move in the direction opposite to the direction of the arrow M1 in response to the movement of the nut **181** in the direction opposite to the direction of the arrow M1. As a result, the cylinder body **103** contracts.

(175) As described above, the electric cylinder **100** is configured such that the cylinder body **103** expands and contracts in response to the forward and reverse rotation of the motor **101**.

(176) <Flow of Lubricant>

(177) Hereinafter, an exemplary example of a flow of the lubricant will be described.

(178) As shown in FIG. **18**, first, the first lid member **165** is removed from the cover unit **160** to open the first supply opening **161a**. Then, the grease nipple **136** is exposed through the first supply opening **161a**. Next, the grease nipple **136** is opened by applying pressure to the grease nipple **136** by connecting a grease gun or the like, and the lubricant is supplied to an inner peripheral side (gap) of the transfer gear **121** through the supply hole **135a** (in a direction of an arrow L1 in the drawing). Then, the lubricant enters an inner peripheral side (gap) of the carriers **114** and **115** along an outer periphery (gap of the spline) of the transfer shaft **122** (in a direction of an arrow L2 in the drawing). Thereafter, the lubricant enters the hollow portion **111a** of the sun gear **111** (in a direction of an arrow L3 in the drawing). As a result, the lubricant can be stored in the hollow portion **111a**.

(179) In a case in which the output shaft **105** rotates in response to the driving of the motor **101**, the sun gear **111** rotates. Then, the lubricant in the hollow portion **111a** of the sun gear **111** flows from the axial distal end portion of the sun gear **111** toward the outer side in the radial direction due to the centrifugal force. Then, a part of the lubricant that has come out from the axial distal end portion of the sun gear **111** flows in a direction of an arrow LA along the first guide groove **143a** of the first carrier **114**, and enters an inner peripheral side (gap) of the planetary gear **112** and a side surface side (gap) of the planetary gear **112**. Thereafter, the lubricant enters an inner peripheral side (gap) of the ring gear **116**. As a result, the sun gear **111**, the planetary gear **112**, and the ring gear **116** can be lubricated.

(180) On the other hand, another part of the lubricant that has come out of the axial distal end portion of the sun gear **111** flows along the outer periphery of the sun gear **111** in a direction of an arrow L5, and flows toward the capturing portion **216**. A part of the lubricant that flows toward the capturing portion **216** flows in a direction of an arrow L6 along the axial distal end surface of the capturing portion **216**, and flows toward the inner peripheral surface of the outer recess portion **118c**. Thereafter, the lubricant enters the inner peripheral side (gap) of the ring gear **116** along the inner peripheral surface of the outer recess portion **118c**. As a result, the sun gear **111**, the planetary gear **112**, and the ring gear **116** can be lubricated.

(181) A part of the lubricant that flows along the outer periphery of the sun gear **111** flows in a direction of an arrow L7 along the second guide groove **151a** of the second carrier **115**, and enters the inner peripheral side (gap) of the planetary gear **112** and the side surface side (gap) of the planetary gear **112**. Thereafter, the lubricant enters an inner peripheral side (gap) of the ring gear **116**. As a result, the sun gear **111**, the planetary gear **112**, and the ring gear **116** can be lubricated.

(182) It should be noted that a part of the lubricant that flows along the outer periphery of the sun gear **111** may enter the inner recess portion **118b** of the holder **118**. As a result, the lubricant can be stored in the inner recess portion **118b** of the holder **118**.

(183) As described above, for example, the grease nipple **136** is opened by the grease gun or the like, the lubricant is supplied through the supply hole **135a**, and the motor **101** is driven, so that the lubricant intervenes between the hollow portion **111a** of the sun gear **111**, the meshing portion of the gears, the gap between the motor **101** and the holder **118**, and the like. As a result, the heat generated by the motor **101** and the heat generated by the friction of each portion can be released to the outside through the portion with the intervening lubricant as a path. Therefore, cooling the motor **101** and the planetary gear mechanism **110** can be promoted.

(184) <Actions and Effects>

(185) As described above, the electric cylinder **100** according to the present embodiment includes

the motor **101** that is a driving source, the output shaft **105** configured to rotate in response to the driving of the motor **101**, the sun gear **111** coupled to the output shaft **105** and configured to rotate in response to the rotation of the output shaft **105**, the tubular holder **118** disposed adjacent to the axial end surface **101f** of the motor **101** and including the fluid accommodation portion **201** that allows the magnetic fluid **200** to be accommodated between the holder **118** and the outer periphery of the sun gear **111**, the pair of magnetic pole pieces **211** and **212** provided on the inner peripheral surface of the holder **118**, and the magnet **210** disposed between the pair of magnetic pole pieces **211** and **212** and configured to hold the magnetic fluid **200** in the fluid accommodation portion **201** by using the action of the magnetic field.

(186) In this configuration, the magnetic fluid **200** can be held in the fluid accommodation portion **201** by using the action of the magnetic field. As a result, the gap between the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the holder **118** is filled with the magnetic fluid **200**. Therefore, a heat transfer path from the output shaft **105** to the holder **118** and a heat transfer path from the tooth surface of the sun gear **111** to the holder **118** can be maintained via the magnetic fluid **200**. Therefore, in a case in which the heat generated by the motor **101** is transferred from the output shaft **105** to the sun gear **111**, the heat can be released to the outside through the portion with the intervening magnetic fluid **200**. In addition, the heat generated on the tooth surface of the sun gear **111** can be released to the outside through the portion with the intervening magnetic fluid **200**. Therefore, the heat from the motor **101** and the heat from the tooth surface of the sun gear **111** can be efficiently released to the outside.

(187) For example, in a case in which the lubricant, such as grease, intervenes in the above-described gap, the lubricant is scattered due to the rotation of the sun gear **111** or the like, so that it is difficult to maintain the heat transfer path from the output shaft **105** to the holder **118**. On the other hand, in the present embodiment, even in a case in which the sun gear **111** rotates, the magnetic fluid **200** can be held in the fluid accommodation portion **201** by using the action of the magnetic field. In addition, even in a case in which the temperature is increased, the magnetic fluid **200** is held in the fluid accommodation portion **201** by using the action of the magnetic field, so that the heat transfer path can be maintained.

(188) In the present embodiment, the output shaft **105** protrudes outward in the axial direction from the axial end surface **101f** of the motor **101**. The sun gear **111** is formed in a tubular shape coaxial with the output shaft **105**. The magnetic pole pieces **211** and **212** are formed in an annular shape coaxial with the sun gear **111**. The fluid accommodation portion **201** includes the gaps **201a** and **201b** that allow the magnetic fluid **200** to be accommodated between the outer peripheral surface of the sun gear **111** and the inner peripheral surfaces of the magnetic pole pieces **211** and **212**.

(189) In this configuration, the gaps **201a** and **201b** are formed in an annular shape along the outer peripheral surface of the sun gear **111** and the inner peripheral surfaces of the magnetic pole pieces **211** and **212**. The magnetic fluid **200** is held in the annular gaps **201a** and **201b** formed in a ring shape by using the action of the magnetic field. Therefore, the heat transfer path from the output shaft **105** to the holder **118** and the heat transfer path from the tooth surface of the sun gear **111** to the holder **118** can be maintained via the magnetic fluid **200** held in the annular gaps **201a** and **201b**. Therefore, in a case in which the heat generated by the motor **101** is transferred from the output shaft **105** to the sun gear **111**, the heat can be released to the outside through the portion with the intervening magnetic fluid **200** between the annular gaps **201a** and **201b** and the inner peripheral surfaces of the magnetic pole pieces **211** and **212**. In addition, the heat generated on the tooth surface of the sun gear **111** can be released to the outside through the portion with the intervening magnetic fluid **200** between the annular gaps **201a** and **201b** and the inner peripheral surfaces of the magnetic pole pieces **211** and **212**. Therefore, the heat from the motor **101** and the heat from the tooth surface of the sun gear **111** can be further efficiently released to the outside.

(190) In the present embodiment, the magnet **210** is formed in an annular shape coaxial with the sun gear **111**. The fluid accommodation portion **201** includes the space **201c** that allows the

magnetic fluid **200** to be accommodated between the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the magnet **210**.

(191) In this configuration, the space **201c** is formed in an annular shape along the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the magnet **210**. The magnetic fluid **200** is held in the space **201c** formed in an annular shape by using the action of the magnetic field. Therefore, the heat transfer path from the output shaft **105** to the holder **118** and the heat transfer path from the tooth surface of the sun gear **111** to the holder **118** can be maintained via the magnetic fluid **200** held in the annular space **201c**. Therefore, in a case in which the heat generated by the motor **101** is transferred from the output shaft **105** to the sun gear **111**, the heat can be released to the outside through the portion with the intervening magnetic fluid **200** in the annular space **201c** and the inner peripheral surface of the magnet **210**. In addition, the heat generated on the tooth surface of the sun gear **111** can be released to the outside through the portion with the intervening magnetic fluid **200** in the annular space **201c** and the inner peripheral surface of the magnet **210**. Therefore, the heat from the motor **101** and the heat from the tooth surface of the sun gear **111** can be further efficiently released to the outside.

(192) In the present embodiment, the holder **118** is formed in an annular shape coaxial with the sun gear **111**, and includes the support surface **119a** extending in the radial direction of the sun gear **111**. The pair of magnetic pole pieces **211** and **212** are the first magnetic pole piece **211** disposed between the support surface **119a** and the magnet **210**, and the second magnetic pole piece **212** disposed adjacent to the surface of the magnet **210** opposite to the first magnetic pole piece **211**.

(193) In this configuration, the first gap **201a** is formed in an annular shape along the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the first magnetic pole piece **211**. The second gap **201b** is formed in an annular shape along the outer peripheral surface of the sun gear **111** and the inner peripheral surface of the second magnetic pole piece **212**. The magnetic fluid **200** is held in the first gap **201a** and the second gap **201b**, which are formed in an annular shape, by using the action of the magnetic field. Therefore, the heat transfer path from the output shaft **105** to the holder **118** and the heat transfer path from the tooth surface of the sun gear **111** to the holder **118** can be maintained via the magnetic fluid **200** held in the first gap **201a** and the second gap **201b**, which have an annular shape. Therefore, in a case in which the heat generated by the motor **101** is transferred from the output shaft **105** to the sun gear **111**, the heat can be released to the outside through the portion with the intervening magnetic fluid **200** between the annular first gap **201a** and the second gap **201b**, the inner peripheral surfaces of the first magnetic pole piece **211** and the second magnetic pole piece **212**. In addition, the heat generated on the tooth surface of the sun gear **111** can be released to the outside through the portion with the intervening magnetic fluid **200** between the first and second gaps **201a** and **201b**, which have an annular shape, and the inner peripheral surfaces of the first magnetic pole piece **211** and the second magnetic pole piece **212**. Therefore, the heat from the motor **101** and the heat from the tooth surface of the sun gear **111** can be further efficiently released to the outside.

(194) For example, as shown in FIG. **19**, the heat from the motor **101** may move along the output shaft **105** in a direction of an arrow H1 and move along the coupling tubular portion **111b** of the sun gear **111** in a direction of an arrow H2. In this case, the heat of the coupling tubular portion **111b** of the sun gear **111** moves in a direction of an arrow H3 through the portion with the intervening magnetic fluid **200** in the annular first gap **201a** and the inner peripheral surface of the first magnetic pole piece **211**. The heat moved in the direction of the arrow H3 is toward the outside through the holder **118** and the case **106**. Therefore, the heat from the motor **101** can be efficiently released to the outside. For example, as shown in FIG. **19**, the heat generated on the tooth surface of the sun gear **111** may move along the coupling tubular portion **111b** of the sun gear **111** in a direction of an arrow H4. In this case, the heat of the coupling tubular portion **111b** of the sun gear **111** moves in a direction of an arrow H5 through the portion with the intervening magnetic fluid **200** in the annular second gap **201b** and the inner peripheral surface of the second magnetic

pole piece **212**. The heat moved in the direction of the arrow H5 is toward the outside through the holder **118** and the case **106**. Therefore, the heat of the tooth surface of the sun gear **111** can be efficiently released to the outside.

(195) In the present embodiment, the electric cylinder **100** includes the retaining ring **215** disposed adjacent to the surface of the second magnetic pole piece **212** opposite to the magnet **210** and configured to stop the second magnetic pole piece **212** from coming off outward in the axial direction of the motor **101**. The holder **118** includes the fitting groove **119b** recessed to allow the retaining ring **215** to be fitted.

(196) In this configuration, the retaining ring **215** is fitted into the fitting groove **119b** of the holder **118**, whereby the second magnetic pole piece **212** can be stopped from coming out outward in the axial direction of the motor **101**. In addition, the number of components can be reduced and the cost can be reduced as compared with a case in which the fitting groove **119b** is provided in a member other than the holder **118**.

(197) In the present embodiment, the holder **118** includes the outer recess portion **118c** that is open outward in the axial direction of the motor **101**. The inner peripheral surface of the outer recess portion **118c** is disposed on the axial outer side of the motor **101** with respect to the fluid accommodation portion **201** and is disposed on the radial outer side of the sun gear **111** with respect to the fluid accommodation portion **201**. The electric cylinder **100** includes the capturing portion **216** that protrudes from the outer peripheral surface of the sun gear **111** toward the inner peripheral surface of the outer recess portion **118c**. The capturing portion **216** can capture the lubricant attempting to flow from the axial distal end portion side of the sun gear **111** toward the fluid accommodation portion **201**.

(198) In this configuration, the lubricant attempting to flow from the axial distal end portion side of the sun gear **111** toward the fluid accommodation portion **201** can be captured by the capturing portion **216**, so that the entry of the lubricant into the fluid accommodation portion **201** can be suppressed. Therefore, it is possible to suppress the lubricant from being mixed with the magnetic fluid **200** held in the fluid accommodation portion **201**. In addition, in a case in which the sun gear **111** rotates, the lubricant flowing toward the capturing portion **201** flows toward the inner peripheral surface of the outer recess portion **118c** due to the centrifugal force. Therefore, the lubricant can be supplied to the inner peripheral surface of the outer recess portion **118c**. For example, as shown in FIG. **18**, another part of the lubricant that has come out from the axial distal end portion of the sun gear **111** flows along the outer periphery of the sun gear **111** in the direction of the arrow L5 and flows toward the capturing portion **216**. A part of the lubricant that flows toward the capturing portion **216** flows in the direction of the arrow L6 along the axial distal end surface of the capturing portion **216** due to the centrifugal force, and flows toward the inner peripheral surface of the outer recess portion **118c**. Thereafter, the lubricant enters the inner peripheral side (gap) of the ring gear **116** along the inner peripheral surface of the outer recess portion **118c**. As a result, the sun gear **111**, the planetary gear **112**, and the ring gear **116** can be lubricated.

(199) In the present embodiment, the excavator **1** includes the vehicle body **2**, and the work equipment **3** coupled to the vehicle body **2**. The work equipment **3** includes the electric cylinder **100**.

(200) Therefore, it is possible to provide the excavator **1** that can efficiently release the heat from the motor **101** and the heat from the tooth surface of the sun gear to the outside.

(201) In the present embodiment, the work equipment **3** includes the common electric cylinder **100** as the first electric cylinder **100A**, the second electric cylinder **100B**, and the third electric cylinder **100C**.

(202) Therefore, the number of components can be reduced and the cost can be reduced as compared with a case in which electric cylinders different from each other are provided as the first electric cylinder **100A**, the second electric cylinder **100B**, and the third electric cylinder **100C**.

OTHER EMBODIMENTS

(203) In the above-described embodiment, the exemplary example has been described in which the electric cylinder includes the planetary gear mechanism that transmits the driving force of the motor to the piston, but the present invention is not limited to this. For example, the electric cylinder need not include the planetary gear mechanism. For example, the electric cylinder may include a power transmission mechanism other than the planetary gear mechanism, such as a belt pulley mechanism or a rack and pinion mechanism. For example, the aspect of the power transmission mechanism can be changed depending on the required specifications.

(204) In the above-described embodiment, the exemplary example has been described in which the electric cylinder includes the sun gear that rotates in response to the rotation of the output shaft, the planetary gear that rotates in response to the rotation of the sun gear, and the tubular holder including the fluid accommodation portion that can accommodate the magnetic fluid between the holder and the outer periphery of the sun gear, but the present invention is not limited to this. For example, the electric cylinder may include a pulley that rotates in response to the rotation of the output shaft, a belt that rotates in response to the rotation of the pulley, and a tubular holder having a fluid accommodation portion capable of accommodating the magnetic fluid between the holder and an outer periphery of the pulley. For example, the electric cylinder may include a pinion that rotates in response to the rotation of the output shaft, a belt that moves in response to the rotation of the pinion, a gear that rotates in response to the movement of the belt, and a tubular holder having a fluid accommodation portion capable of accommodating the magnetic fluid between the holder and an outer periphery of the pinion. For example, the electric cylinder need only include a motor that is a driving source, an output shaft configured to rotate in response to driving of the motor, a rotating body coupled to the output shaft and configured to rotate in response to the rotation of the output shaft, a tubular holder disposed adjacent to an axial end surface of the motor and including a fluid accommodation portion that allows a magnetic fluid to be accommodated between the holder and an outer periphery of the rotating body, a pair of magnetic pole pieces provided on an inner peripheral surface of the holder, and a magnet disposed between the pair of magnetic pole pieces and configured to hold the magnetic fluid in the fluid accommodation portion by using an action of a magnetic field. For example, the aspect of the rotating body that rotates in response to the rotation of the output shaft can be changed depending on the required specifications.

(205) In the above-described embodiment, the exemplary example has been described in which the magnetic pole piece is formed in an annular shape coaxial with the sun gear, but the present invention is not limited to this. For example, the magnetic pole piece need not be formed in an annular shape coaxial with the sun gear. For example, the magnetic pole piece may be formed by disposing a plurality of small magnetic pole pieces to be arranged in an annular shape. For example, the aspect of the magnetic pole piece can be changed depending on the required specifications.

(206) In the above-described embodiment, the exemplary example has been described in which the magnet is formed in an annular shape coaxial with the sun gear, but the present invention is not limited to this. For example, the magnet need not be formed in an annular shape coaxial with the sun gear. For example, the magnet may be formed by disposing a plurality of small magnets to be arranged in an annular shape. For example, the aspect of the magnet can be changed depending on the required specifications.

(207) In the above-described embodiment, the exemplary example has been described in which the fluid accommodation portion includes the space that allows the magnetic fluid to be accommodated between the outer peripheral surface of the sun gear and the inner peripheral surface of the magnet, but the present invention is not limited to this. For example, the fluid accommodation portion need not include the space that allows the magnetic fluid to be accommodated between the outer peripheral surface of the sun gear and the inner peripheral surface of the magnet. For example, the fluid accommodation portion need only include a gap that allows the magnetic fluid to be

accommodated between the outer peripheral surface of the sun gear and the inner peripheral surface of the magnetic pole piece. For example, the aspect of the fluid accommodation portion can be changed depending on the required specifications.

(208) In the above-described embodiment, the exemplary example has been described in which the inner peripheral surface of the magnet is disposed on a radial outer side of the sun gear with respect to the inner peripheral surface of the magnetic pole piece, but the present invention is not limited to this. For example, the inner peripheral surface of the magnet need not be disposed on the radial outer side of the sun gear with respect to the inner peripheral surface of the magnetic pole piece. For example, the inner peripheral surface of the magnet may be disposed on the radial inner side of the sun gear with respect to the inner peripheral surface of the magnetic pole piece. For example, the inner peripheral surface of the magnet may be disposed at the same position as the inner peripheral surface of the magnetic pole piece in the radial direction of the sun gear. For example, the disposition aspect of the inner peripheral surface of the magnet can be changed depending on the required specifications.

(209) In the above-described embodiment, the exemplary example has been described in which the holder is formed in an annular shape coaxial with the sun gear and includes the support surface extending in the radial direction of the sun gear, but the present invention is not limited to this. For example, the holder need not include the support surface. For example, the support surface may be provided on a member other than the holder. For example, the aspect of the holder can be changed depending on the required specifications.

(210) In the above-described embodiment, the exemplary example has been described in which the pair of magnetic pole pieces are the first magnetic pole piece disposed between the support surface and the magnet, and the second magnetic pole piece disposed adjacent to the surface of the magnet opposite to the first magnetic pole piece, the magnet, the first magnetic pole piece, and the second magnetic pole piece constitute the fluid holding structure for holding the magnetic fluid in the fluid accommodation portion, and only one fluid holding structure is provided on the inner peripheral surface of the accommodation recess portion, but the present invention is not limited to this. For example, a plurality of the fluid holding structures may be provided on the inner peripheral surface of the accommodation recess portion of the holder. For example, two or more magnets and three or more magnetic pole pieces may be alternately disposed adjacent to each other. For example, the installation aspects of the magnet, the first magnetic pole piece, and the second magnetic pole piece (installation aspect of the fluid holding structure) can be changed depending on the required specifications.

(211) In the above-described embodiment, the exemplary example has been described in which the electric cylinder includes the retaining ring disposed adjacent to the surface of the second magnetic pole piece opposite to the magnet and stops the second magnetic pole piece from coming off outward in an axial direction of the motor, but the present invention is not limited to this. For example, the electric cylinder need not include the retaining ring. For example, the installation aspect of the retaining ring can be changed depending on the required specifications. For example, the outer peripheral portion of the magnetic pole piece may be fixed by the screwing onto the inner peripheral surface of the accommodation recess portion of the holder. For example, the magnetic pole piece may be fixed via a plate and a bolt. For example, the holding aspect of the magnetic pole piece and the magnet can be changed depending on the required specifications.

(212) In the above-described embodiment, the exemplary example has been described in which the holder includes the fitting groove that is recessed to allow the retaining ring to be fitted, but the present invention is not limited to this. For example, the holder need not include the fitting groove. For example, the fitting groove may be provided in a member other than the holder. For example, the installation aspect of the fitting groove can be changed depending on the required specifications.

(213) In the above-described embodiment, the exemplary example has been described in which the

electric cylinder includes the capturing portion that protrudes from the outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion of the holder and can capture the lubricant attempting to flow from the axial distal end portion side of the sun gear toward the fluid accommodation portion, but the present invention is not limited to this. For example, the electric cylinder need not include the capturing portion. For example, the installation aspect of the capturing portion can be changed depending on the required specifications.

(214) In the above-described embodiment, the exemplary example has been described in which the work equipment includes the common electric cylinder as the first electric cylinder, the second electric cylinder, and the third electric cylinder, but the present invention is not limited to this. For example, the work equipment may include electric cylinders different from each other as the first electric cylinder, the second electric cylinder, and the third electric cylinder. For example, the installation aspect of the electric cylinder can be changed depending on the required specifications.

(215) In the above-described embodiment, the excavator has been described as an exemplary example of the work machine (work vehicle), but the present invention is not limited to this. For example, the present invention may be applied to other work vehicles, such as a dump truck, a bulldozer, and a wheel loader.

(216) Although the embodiments of the present invention have been described above, the present invention is not limited to this, additions, omissions, substitutions, and other modifications of the configuration can be made without departing from the spirit of the present invention, and the above-described embodiments can be appropriately combined.

REFERENCE SIGNS LIST

(217) **1**: Excavator (work machine) **2**: Vehicle body **3**: Work equipment **100**: Electric cylinder **100A**: First electric cylinder (electric cylinder) **100B**: Second electric cylinder (electric cylinder) **100C**: Third electric cylinder (electric cylinder) **101**: Motor **101f**: Axial end surface of motor **101A**: First motor (motor) **101B**: Second motor (motor) **101C**: Third motor (motor) **105**: Output shaft **111**: Sun gear **118**: Holder **118c**: Outer recess portion **119a**: Support surface **119b**: Fitting groove **200**: Magnetic fluid **201**: Fluid accommodation portion **201a**: First gap (gap) **201b**: Second gap (gap) **201c**: Space **210**: Magnet **211**: First magnetic pole piece (magnetic pole piece) **212**: Second magnetic pole piece (magnetic pole piece) **215**: Retaining ring **216**: Capturing portion

Claims

1. An electric cylinder comprising: a motor that is a driving source; an output shaft configured to rotate in response to driving of the motor; a sun gear coupled to the output shaft and configured to rotate in response to rotation of the output shaft; a tubular holder disposed adjacent to an axial end surface of the motor and including a fluid accommodation portion that allows a magnetic fluid to be accommodated between the holder and an outer periphery of the sun gear; a pair of magnetic pole pieces provided on an inner peripheral surface of the holder; and a magnet disposed between the pair of magnetic pole pieces and configured to hold the magnetic fluid in the fluid accommodation portion by using an action of a magnetic field.
2. The electric cylinder according to claim 1, wherein the output shaft protrudes outward in an axial direction from the axial end surface of the motor, the sun gear is formed in a tubular shape coaxial with the output shaft, the magnetic pole piece is formed in an annular shape coaxial with the sun gear, and the fluid accommodation portion includes a gap that allows the magnetic fluid to be accommodated between an outer peripheral surface of the sun gear and an inner peripheral surface of the magnetic pole piece.
3. The electric cylinder according to claim 2, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the

fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

4. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 2.

5. The electric cylinder according to claim 2, wherein the magnet is formed in an annular shape coaxial with the sun gear, and the fluid accommodation portion further includes a space that allows the magnetic fluid to be accommodated between the outer peripheral surface of the sun gear and an inner peripheral surface of the magnet.

6. The electric cylinder according to claim 5, wherein the holder is formed in an annular shape coaxial with the sun gear and includes a support surface extending in a radial direction of the sun gear, and the pair of magnetic pole pieces include a first magnetic pole piece disposed between the support surface and the magnet, and a second magnetic pole piece disposed adjacent to a surface of the magnet opposite to the first magnetic pole piece.

7. The electric cylinder according to claim 5, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

8. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 5.

9. The electric cylinder according to claim 5, wherein the inner peripheral surface of the magnet is disposed on a radial outer side of the sun gear with respect to the inner peripheral surface of the magnetic pole piece.

10. The electric cylinder according to claim 9, wherein the holder is formed in an annular shape coaxial with the sun gear and includes a support surface extending in a radial direction of the sun gear, and the pair of magnetic pole pieces include a first magnetic pole piece disposed between the support surface and the magnet, and a second magnetic pole piece disposed adjacent to a surface of the magnet opposite to the first magnetic pole piece.

11. The electric cylinder according to claim 9, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

12. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 9.

13. The electric cylinder according to claim 2, wherein the holder is formed in an annular shape coaxial with the sun gear and includes a support surface extending in a radial direction of the sun gear, and the pair of magnetic pole pieces include a first magnetic pole piece disposed between the support surface and the magnet, and a second magnetic pole piece disposed adjacent to a surface of the magnet opposite to the first magnetic pole piece.

14. The electric cylinder according to claim 13, further comprising: a retaining ring disposed adjacent to a surface of the second magnetic pole piece opposite to the magnet and configured to

stop the second magnetic pole piece from coming off outward in an axial direction of the motor, wherein the holder includes a fitting groove recessed to allow the retaining ring to be fitted therein.

15. The electric cylinder according to claim 14, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

16. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 14.

17. The electric cylinder according to claim 13, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

18. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 13.

19. The electric cylinder according to claim 1, wherein the holder includes an outer recess portion that is open outward in an axial direction of the motor, an inner peripheral surface of the outer recess portion is disposed on an axial outer side of the motor with respect to the fluid accommodation portion and is disposed on a radial outer side of the sun gear with respect to the fluid accommodation portion, and the electric cylinder further comprises a capturing portion protruding from an outer peripheral surface of the sun gear toward the inner peripheral surface of the outer recess portion and configured to capture a lubricant attempting to flow from an axial distal end portion side of the sun gear toward the fluid accommodation portion.

20. A work machine comprising: a vehicle body; and work equipment coupled to the vehicle body, wherein the work equipment includes the electric cylinder according to claim 1.
